

ENTERPRISE NETWORK WITH SECURITY HARDENING

A NOC-Monitored Secure Network - Design, Implementation & Verification

Cisco Packet Tracer 8.2.2 Platform	3 Zones Security Zones	7 Phases Implementation Phases	Completed Project Status
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Target Role	NOC Engineer / Network Security Analyst
Technologies	Cisco ASA 5505 · VLANs · ACLs · NAT/PAT · SSH v2 · Port Security · NTP · Syslog
Key Concept	Three-zone firewall architecture (Internet → DMZ → LAN) with defense-in-depth

1. PROJECT OVERVIEW

This project simulates a real-world, security-hardened enterprise network using Cisco Packet Tracer 8.2.2. A Cisco ASA 5505 firewall divides the network into three distinct security zones, enforcing strict traffic policies that mirror production NOC environments. Every design decision follows the principle of least privilege herby ensuring that no zone can access another unless explicitly permitted.

What Problem Does This Solve?

Organizations that expose public services (web, email) directly on their internal network face catastrophic risk if those servers are compromised. This architecture solves that by creating an isolated DMZ - a semi-trusted buffer zone, so that even a fully compromised public server cannot reach internal systems.

2. SECURITY ZONE ARCHITECTURE

Zone	Security Level	What Lives Here	The Key Rule
OUTSIDE - Internet	0 - Zero Trust	ISP Router, External PCs	All traffic blocked unless explicitly permitted by ACL
DMZ - Demilitarized	50 - Semi-Trusted	WEB-SERVER, MAIL-SERVER, NOC-SERVER	Internet can reach services; DMZ cannot initiate to LAN
INSIDE - LAN	100 - Fully Trusted	HR, Finance, IT, Management PCs	Can reach DMZ and internet; no inbound from lower zones

The ASA firewall enforces these levels automatically - higher-to-lower traffic is allowed; lower-to-higher is blocked by default. A critical design detail: the no forward interface vlan 1 command on the ASA 5505 physically prevents DMZ servers from initiating any connection to the LAN, even if the firewall were misconfigured.

Topology at a Glance

INTERNET ZONE ISP-RT • OUTSIDE-PC	DMZ (172.16.10.0/24) WEB • MAIL • NOC-SERVER	LAN INSIDE HR • Finance • IT • Mgmt
Sec Level: 0	ASA-FW enforces boundary	Sec Level: 100

3. NETWORK DESIGN & IP ADDRESSING

IP Scheme - Clean Separation by Zone

Zone / Segment	Subnet	Gateway	Key Devices
Internet WAN Link	203.0.113.0/30	203.0.113.1 (ASA)	ASA-FW ↔ ISP-RT
DMZ	172.16.10.0/24	172.16.10.1 (ASA)	WEB, MAIL, NOC servers
ASA ↔ LAN L3SW Link	192.168.100.0/30	192.168.100.1 (ASA)	ASA-FW ↔ LAN-L3SW
LAN VLAN 10 - HR	192.168.10.0/24	192.168.10.1	PC-HR
LAN VLAN 20 - Finance	192.168.20.0/24	192.168.20.1	PC-FINANCE
LAN VLAN 30 - IT	192.168.30.0/24	192.168.30.1	PC-IT
LAN VLAN 40 - Mgmt	192.168.40.0/24	192.168.40.1	PC-MGT

ASA Firewall - Core Configuration Highlights

Three named interfaces with enforced security levels. NAT/PAT translates both LAN and DMZ traffic to the ASA public IP for outbound internet access. An inbound ACL on the outside interface selectively allows only the ports needed for public services:

```
! Three security zones on ASA 5505
interface vlan 1 → nameif inside → security-level 100 → 192.168.100.1/24
interface vlan 2 → nameif outside → security-level 0 → 203.0.113.1/30
interface vlan 3 → nameif dmz → security-level 50 → 172.16.10.1/24
no forward interface vlan 1 ! DMZ cannot reach LAN

! PAT - both LAN and DMZ share ASA public IP for outbound
nat (inside,outside) dynamic interface
nat (dmz,outside) dynamic interface

! Only allow what is needed from internet into DMZ
permit tcp any host 172.16.10.10 eq 80 ! HTTP → WEB-SERVER
permit tcp any host 172.16.10.10 eq 443 ! HTTPS → WEB-SERVER
permit tcp any host 172.16.10.20 eq 25 ! SMTP → MAIL-SERVER
permit udp host 203.0.113.2 host 172.16.10.30 eq 514 ! Syslog → NOC
```

LAN Segmentation - VLANs & ACLs

The Cisco 3650 Layer 3 switch handles inter-VLAN routing via SVIs. A single named ACL (LAN-SECURITY) applied inbound on every VLAN SVI enforces zero-trust between departments:

Rule	Traffic	Effect
1. IT Full Access	192.168.30.0/24 → any	IT department has unrestricted access
2. Finance → HR	192.168.20.0/24 → 192.168.10.0/24	DENY - Finance cannot probe HR subnet
3. HR → Finance	192.168.10.0/24 → 192.168.20.0/24	DENY - HR cannot probe Finance subnet
4. HR/Finance → DMZ	192.168.10-20.0/24 → 172.16.10.0/24	DENY - no direct DMZ access
5. Implicit Permit	any → any	All other traffic allowed

4. DEVICE HARDENING & MONITORING

Beyond the firewall, every device in the network was hardened individually - eliminating attack surface that default configurations leave open.

SSH v2 - Encrypted Management (Telnet Eliminated)

All routers and switches enforce SSH-only access. Local credentials, RSA key encryption, limited retries, and automatic idle session timeout prevent unauthorized remote sessions.

```
ip domain-name corporate.local
crypto key generate rsa modulus 1024
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
line vty 0 4
  transport input ssh    ! Telnet completely disabled
  login local
  exec-timeout 5 0      ! Auto-disconnect idle sessions after 5 min
  no cdp run            ! Disable Cisco Discovery Protocol
  service password-encryption
```

Port Security - MAC-Based Access Control

Access ports on LAN-SW1, LAN-SW2 and DMZ-SW are locked to a single MAC address using sticky learning. The restrict mode was chosen deliberately: the port stays active, violations are dropped silently, and every event is logged to the NOC Syslog server for investigation.

Violation Mode	Port Stays Up?	Logs to Syslog?	Why Chosen?
Protect	Yes	No	Too silent - NOC cannot see attacks
Restrict	Yes	Yes	Selected - port stays up, NOC gets full visibility
Shutdown	No - err-disabled	Yes	Too disruptive - legitimate users lose access

NTP + Syslog - Centralized Time & Logging

NOC-SERVER (172.16.10.30) acts as both the NTP master and Syslog collector. NTP must be configured before Syslog because timestamps on log events are meaningless if devices disagree on the time.

Component	Configuration	Purpose
NTP Master	NOC-SERVER - stratum 1 source	Single time reference for all devices
NTP Clients	ntp server 172.16.10.30 on all IOS devices	All clocks synchronized
Syslog Level	logging trap informational (severity 0–6)	All significant events captured; debug excluded
Syslog Target	logging host 172.16.10.30	All logs centralized on NOC-SERVER
Timestamps	service timestamps log datetime msec	Millisecond precision for incident forensics

Design Insight: NTP sync on Cisco 2960 switches fails in Packet Tracer 8.2.2 due to a known simulator bug I confirmed, not a configuration error. The applied configuration is correct per Cisco IOS documentation and would sync successfully on real hardware.

5. VERIFICATION & TEST RESULTS

All configuration phases were verified end-to-end. Results below confirm the network behaves exactly as designed.

Firewall Zone Tests

Test	From	To	Expected	Result
Internet → WEB (HTTP)	OUTSIDE-PC	172.16.10.10:80	ALLOWED (ACL rule)	PASS
Internet → LAN	OUTSIDE-PC	192.168.10.10	BLOCKED (firewall)	PASS
DMZ → LAN initiation	WEB-SERVER	192.168.10.10	BLOCKED (no forward)	PASS
LAN-IT → DMZ	PC-IT	172.16.10.30	ALLOWED	PASS
LAN → Internet	PC-FINANCE	8.8.8.10	ALLOWED + NATted	PASS

ACL Department Isolation Tests

Test	Source	Destination	Result
Finance cannot reach HR	PC-FINANCE (VLAN 20)	PC-HR (192.168.10.10)	BLOCKED
HR cannot reach Finance	PC-HR (VLAN 10)	PC-FINANCE (192.168.20.10)	BLOCKED
Finance can reach IT	PC-FINANCE	PC-IT (192.168.30.10)	ALLOWED
IT can reach DMZ	PC-IT	172.16.10.10 (WEB-SRV)	ALLOWED
HR/Finance blocked from DMZ	PC-HR / PC-FINANCE	172.16.10.10	BLOCKED

6. KEY LESSONS & TROUBLESHOOTING

Issue Encountered	Root Cause	Resolution
ASA nameif error on Eth ports	ASA 5505 physical ports are Layer 2 only	Configure nameif on VLAN interfaces, not physical ports
3rd ASA interface rejected	Base license limits to 2 named interfaces	Used no forward interface vlan 1 to unlock DMZ interface
DHCP conflict on ASA startup	Default DHCP pool occupied the inside subnet	Removed default pool with no dhcpd enable inside first
Syslog missing from SW1/SW2	Management IPs not routable back through ASA	Understood as return-path failure; documented for prod fix
ISP-RT syslog dropped at ASA	UDP 514 not in OUTSIDE-TO-DMZ ACL	ACL updated to permit UDP 514 from ISP-RT to NOC-SERVER

Skills Demonstrated

Firewall Architecture ASA zone design, security levels, stateful inspection, NAT/PAT	DMZ Design 3-zone separation, public service exposure, no-forward restriction	Network Security Least privilege, defense-in-depth, zone + port-based filtering
Device Hardening SSH v2, RSA keys, exec-timeout, CDP off, HTTP off	Port Security MAC limiting, sticky learning, restrict mode, syslog verification	Monitoring & Logging Syslog severity levels, NTP hierarchy, timestamp forensics

This project was built from scratch in Cisco Packet Tracer 8.2.2 as an entry-level IT security portfolio piece. It demonstrates practical competence in firewall architecture, device hardening, access control and NOC monitoring - directly applicable to NOC Engineer, Network Security Analyst and Junior Network Engineer roles.